<u>AEOLUS AVIONICS ROUND 2 – TASK 1</u>

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Build 2: Depth camera (OAK-D LITE + AI) with Raspberry Pi 5 8 GB RAM.

Compute: Uses Raspberry Pi 5 which gives solid general purpose computing power but slower than the Jetson.

Power: 6.6 – 17.6W

Weight: <121-131gms

Cost: ₹56,748

Priority Order: Use case > Weight > Power > Compute > Cost

For the **USE CASE**, we need a drone with good endurance and sensors and flight computer which are light weight since the weight is limited to 2.5kg and 10-inch frame. It must also be able to run person-detection. Build 1 has a more powerful computer (NVIDIA Jetson Orin Nano) which has 6 core ARM cortex A78E CPU, 1024 CUDA core Ampere GPU, 40 TOPS AI performance. But I feel that much performance is not worth sacrificing endurance which is very important for a SAR mission over the ocean. Additionally, Build 2 makes up for the lack of AI on the Raspberry by having it on the OAK-D, which also has the highest RGB resolution in the list. It is to be noted that the depth-sensing capabilities of the OAK-D are more or less useless over a sunny ocean so we'll have to rely on the RGB camera. Though the compute is still no match for the Jetson, I think it's enough for the mission we have and worth sacrificing the extra compute for more endurance & lesser power consumption on board. Compared to Build 1, it's cheaper, lighter and consumes lesser power. Thus, giving the drone more endurance in comparison to the first build. OAK-D's own AI silicon can run person-detection inference independently of the flight computer.

Build 5 is both the most expensive and heavy and doesn't fit the use case well, ie, the D435i's depth sensing doesn't help with wide-area detection any more than Build 1's stereo camera does, and it adds no AI acceleration the Jetson doesn't already provide on its own. Build 4 on the other hand is the exact opposite – cheapest, lightest and least power consumption. Though, it doesn't have onboard AI and the passive stereo camera with narrow FOV make it a poor fit for the use case. Build 3 again beats Build 2 on power, weight and cost but like Build 4, it falls short of the use case which is our highest priority. The LIDAR d500's range (12m) is designed for close range obstacles and not a wide area search and neither do the LIDAR and the Arducam carry any onboard GPU/NPU which means all processing and detection falls on the Pi 5.

BONUS:

The best orientation for the OAK-D is at 45 degrees from the horizontal (or vertical, it's the same since its 45) which maximises the area swept by it.

Though, I must conclude that none of the builds given in the list are ideal for a SAR mission. Even Build 2's OAK-D's depth sensing isn't very useful to the task in hand as mentioned before. Thus, I propose an alternate build: Raspberry Pi 5 8GB RAM + *FLIR Lepton 3.5 (₹29,329)*. This not only removes the AI image recognition part by trying to spot a person visually but also reduces the cost to ₹49,328 and the Lepton doesn't weigh as much as the OAK-D and consumes less power. Since a human's heat signature stands out sharply against the ocean temperature, meaning, a simple blob identification via putting a threshold can be implemented and not the visual identification, which makes the OAK-D's AI unnecessary in this case and Pi 5's CPU is sufficient. Though the limitation is that since I dropped the RGB camera there's no visual confirmation whether the drone has encountered Odysseus or something else which crosses the threshold. But this fits the use case well, has lower weight, consumes less power, and costs less.